



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Software engineering is shaping the future!



Learning Outcome:

Understand information about common solutions to problems in spoken or written documents.
Talk about programming and data base related to hypertext preprocessor in spoken or written documents.

Ask your partner:

- How will software engineering change our lives in 50 years?
- Do you think software engineering will reach a peak it won't be able to surpass, or it could improve infinitely? Why?

Can Do Statements:

- I can read about the predictions on software engineering.
- I can write correctly a text to summarize information related to software programming.
- I can explain my point of view regarding the evolution of software engineering.



Let's Explore!

Some Predictions About What Software Engineering Could Do



Instructions: Read the text below.

How will traffic and transportation be in the future? If I were able to change one thing in our day to day lives, I would choose traffic jams. IBM is about to launch a system to predict traffic jams before they occur. It uses GPS, roadside sensors, and archived data to predict congestion. The technology has been proven successful in trials. If we had realized sooner that cars pollute so much, we could have developed eco-friendly vehicles sooner. However, in the last years, cars have become cleaner, safer, and more affordable, but there is a long way to go. In 2015, Larry Burns, the VP of General Motors predicted that computer-enhanced vehicles that run on hydrogen are possible, and they could even channel energy back into the electrical grid when not in use.

The previous statements are examples of software engineering, a new science that many human lives depend on. In the past few years, newspaper's front pages have been swamped with headlines about the lack of privacy and security of software. It is true that software engineering has created some of the biggest concerns related to credit card theft, identity fraud, and technical malfunctions in cars and airplanes. Due to this, some people argue that software engineering should have never been created. Nevertheless, if the same software is capable of such harm, it can help in numerous ways too. It has already helped to prevent car collisions and improved the way doctors perform surgeries, among other advances.

Let's take a look at apps now; they are invaluable parts of our smartphones. If someone told you that apps would be so far-reaching that they would be able to conduct medical checkups or identify people through face-recognition, would you believe it? Apple and Google are creating face recognition software for cellphones. I think that scientists and engineers from decades ago could have never imagined such developments. Nowadays, they are trying their best to develop apps to measure blood pressure and heart rates. If they manage to create such powerful smartphones, I will definitely buy one!



**Let's Practice!**

Some Relevant Concepts to Keep in Mind!

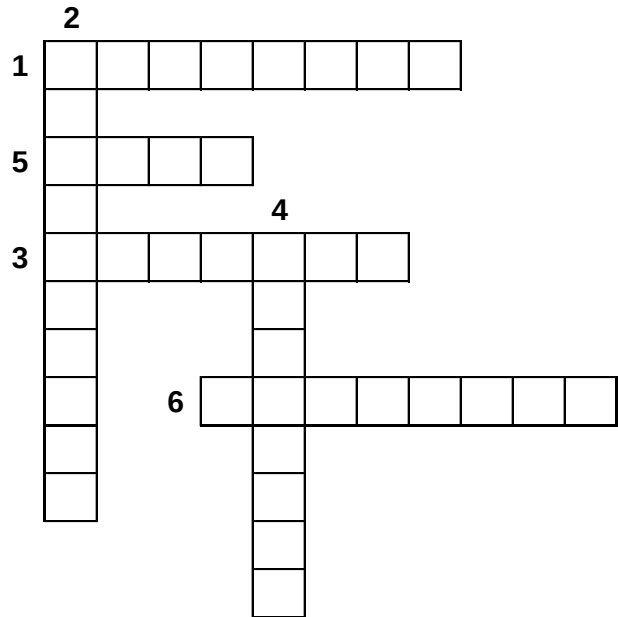
Instructions: Read the clues. Complete the crossword.

DOWN

2. If they manage to create a such powerful _____, I will definitely buy one!
4. If we had realized sooner that cars pollute so much, we could have developed eco-_____ vehicles sooner.

ACROSS

1. If the same _____ is capable of such harm, it can help in numerous ways too.
3. If I were able to change one thing in our day to day lives, I would choose _____ jams.
5. If software programmers hadn't developed so many _____, some of our daily activities wouldn't have been so easy to do.
6. If someone told you that apps would be so far-_____ that they would be able to conduct medical checkups or identify people through face-recognition, would you believe it?

**Let's Practice!**

Learning About Software Programming!

Instructions: Write the words to correctly complete the sentences. Use words from the box below. Read the sentences aloud. Check your pronunciation with your instructor.

should have / should have known / would have been /
would have improved / would have been informed

1. You _____ that many modern inventions depend on software engineering.
2. It could have been better, if people _____ about the lack of privacy and security of software.
3. Citizens _____ concerned about credit card theft, identity fraud, and technical malfunctions in cars and airplanes.
4. If you had researched about the advantages of software applications, you _____ realized that they have prevented car collisions.
5. If engineers had developed medical technology a long time ago, doctors _____ the way of performing life-saving surgeries among numerous other advantages.





Extension Activity

The best is yet to come!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- a) Search for more information about the benefits of software programming.
- b) Make a list of the inventions such as applications and new technologies, you consider have been significant for our lives.
- c) Prepare a presentation about these inventions.
- d) Use the grammar and vocabulary you studied in this learning object.
- e) Present your work in front of your class.
- f) Discuss and answer your classmates and teacher's questions.

Self-Assessment

Yes
No

- | | | |
|---|-----------------------|-----------------------|
| 1. I read about the benefits of software programming. | ☐ | ☐ |
| 2. I list the most important inventions I consider have helped people to improve their quality of life. | ☐ | ☐ |
| 3. I make a presentation explaining each of these applications or new technologies. | ☐ | ☐ |
| 4. I use the grammar and vocabulary I learned. | ☐ | ☐ |
| 5. I talk about the benefits of software programming. | ☐ | ☐ |
| 6. I discuss and answer the questions my classmates and teacher may have. | ☐ | ☐ |





Target Words

- **app** (noun): abbreviation of the word "application", it is a particular program that helps execute tasks into mobile devices, such as cellphones and tablets
e.g. If someone told you that apps would be so far-reaching that they would be able to conduct medical checkups or identify people through face-recognition, would you believe it?
- **computer-enhanced** (adjective): the quality of improving something thanks to computers' technology
e.g. The VP of General Motors predicted that computer-enhanced vehicles that run on hydrogen are possible.
- **far-reaching** (adjective): the quality of having a wide influence over something
e.g. If someone told you that apps would be so far-reaching that they would be able to conduct medical checkups or identify people through face-recognition, would you believe it?
- **launch** (verb): to promote or to release a new product, such as a software program or an application
e.g. IBM is about to launch a system to predict traffic jams before they occur.
- **eco-friendly** (adjective): a product or service that doesn't harm the environment
e.g. If we had realized sooner that cars pollute so much, we could have developed eco-friendly vehicles sooner.
- **channel** (verb): to conduct something into a profitable course
e.g. Larry Burns, the VP of General Motors predicted that computer-enhanced vehicles that run on hydrogen are possible, and they could even channel energy back into the electrical grid when not in use.



Learning Points

Zero, First, Second, and Third Conditional Review

Conditionals refer to any sentence with an -if clause and a -result clause.

We use the **zero conditional** to convey logical conclusions. It is formed by the **present simple** in the if clause and in the result clause.

Example:

***If** some software developments **can** do harm, others **can** do well in the society.*

We use the **first conditional** to talk about the future. If the condition is met then the statement is likely to happen. It is formed by the **present simple** in the if clause, although the **present continuous** is also possible. The result clause is in the future tense. It can have many forms apart from the usual will; going to and present continuous with future meaning are used as well.





Learning Points

Example:

*If they **create** a software capable of translating any language, no one **will** hire a translator anymore.*

We use the **second conditional** to talk about imaginary or hypothetical situations. It is also called the unreal conditional. The verb in the -if clause is in the **past simple** and the verb in the result clause is preceded by the modal verb **would, could** or **might**.

Example:

*If scientists **developed** an app to check your heart rate, many people **would benefit** from it.*

The **third conditional** is used to talk about unreal situations in the past. This situation didn't happen but we're imagining the results. It is formed by the **past perfect** in the *if clause*, and the *result clause* uses **would have** and the **past participle** of the verb.

Example:

*If scientists **had studied** the impact of CO2 that cars produced, engineers **would have created** eco-friendly cars by now.*

Examples

- *If I were able to change one thing in our day to day lives, I would choose traffic jams.*
- *If we had realized sooner that cars pollute so much, we could have developed eco-friendly vehicles sooner.*
- *If the same software is capable of such harm, it can help in numerous ways too.*



Learning Points

Past Modals Review

We use past modals when we want to express past possibilities or imaginary situations that didn't end up as expected.

The most commonly used modals in the past are **should**, **could**, and **would**; to use these modal verbs in the past you must follow this structure:

Subject+ **modal verb** + **have** + **main verb in past participle** + complement

Remember that we use the present perfect with **have** after the modal verb to refer to past expectations.

Should have:

Should is the modal verb we use for giving advice, so use **should have** when you want to mention the best way things should have been done.

Examples:

*You **should have used** the GPS system to find your way.
We **should have bought** an eco-friendly car.*

Could have:

Use **could have** to talk about possibilities in the unreal past; if you want to underline other suitable options if past was different.

Examples:

*I **could have finished** my work earlier, but I have a slow internet connection.
We **could have done** that faster with this new app.*

Would have:

When you want to express imagined final results in the past, use the modal verb **would have**; remember that **would have** expresses more certainty than **could have**.

Examples:

*The manager **would have saved** lots of money with that new software.
You **would have done** that better with a better engine.*

Examples

- *If we had realized sooner that cars pollute so much, we could have developed eco-friendly vehicles sooner.*
- *I think that scientists and engineers from decades ago could have never imagined such developments.*
- *Some people argue that software engineering should have never been created.*





Learning Points

Punctuation and Capitalization

We use different kinds of **punctuation marks** in a text to provide cohesion and coherent. On the other hand, **capitalization** helps us to understand some important words.

The two more used punctuation marks are the **comma** and the **period**.

We use the **period** at the end of a sentence. When there is a question or exclamation mark at the end of a sentence, the **period** is not used. The same happens when there is an abbreviation. The **period** used at the end of the abbreviation also serves as the **period** for the sentence.

Commas are very important punctuation marks as their absence can change the meaning of a sentence. You use the comma in the following cases:

- To list different objects in a sequence. Do not place the comma after the last item of the list.
- If a noun is being modified by a series of adjectives, they should be separated by commas.
- When there is a dependent and independent clause in a sentence, we use the comma to separate them.

Regarding **capitalization**, remember that proper names and languages always have capital letters. On the other hand, the first word at the beginning of a sentence, or after a period is capitalized. Also, abbreviations are completely written in capital letters.

Examples:

Bill Gates
General Motors
The English language
Apple
Google
IBM
GPS

Examples

- *IBM is about to launch a system to predict traffic jams before they occur.*
- *In 2015, Larry Burns, the VP of General Motors predicted that computer-enhanced vehicles that run on hydrogen are possible.*
- *Apple and Google are creating face recognition software for cellphones.*

